

Exercise -1- Create a Spring Web Project using Maven

→https://start.spring.io

☰

🕒

 **spring** initializr

Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ Java ☐ Kotlin ☐ Groovy

☒ Maven

Spring Boot

☐ 4.1.1 (SNAPSHOT) ☒ 4.1.0 ☐ 4.0.8 (SNAPSHOT) ☐ 4.0.7 ☐ 3.5.16

Project Metadata

Group

com.cts

Artifact

SpringRest

Package name

com.cts.SpringRest

Packaging

☒ Jar ☐ War

Configuration

☒ Properties ☐ YAML

Java

☐ 26 ☐ 25 ☐ 21 ☒ 17

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Web WEB

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container. ⊖

Spring Boot DevTools DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience. ⊖

GENERATE CTRL + ⌘

EXPLORE CTRL + SPACE

...

```

32 <dependencies>
33 <dependency>
34 <groupId>org.springframework.boot</groupId>
35 <artifactId>spring-boot-starter-web</artifactId>
36 </dependency>
37
38 <dependency>
39 <groupId>org.springframework.boot</groupId>
40 <artifactId>spring-boot-devtools</artifactId>
41 <scope>runtime</scope>
42 <optional>true</optional>
43 </dependency>
44 <dependency>
45 <groupId>org.springframework.boot</groupId>
46 <artifactId>spring-boot-starter-test</artifactId>
47 <scope>test</scope>
48 </dependency>
49
50 <dependency>
51 <groupId>org.projectlombok</groupId>
52 <artifactId>lombok</artifactId>
53 <version>1.18.46</version>
54 <scope>provided</scope>
55 </dependency>
56 </dependencies>
57

```

```
SpringRest/pom.xml | SpringRestApplication.java X
SpringRest > src/main/java > com.cts.SpringRest > SpringRestApplication > log: Logger
1 package com.cts.SpringRest;
2
3
4 import org.springframework.boot.SpringApplication;
5 import org.springframework.boot.autoconfigure.SpringBootApplication;
6
7 import lombok.extern.slf4j.Slf4j;
8
9 @Slf4j
10 @SpringBootApplication
11 public class SpringRestApplication {
12
13
14 public static void main(String[] args) {
15     log.info("Application is starting...");
16
17     SpringApplication.run(SpringRestApplication.class, args);
18
19     log.info("Application started successfully.");
20 }
21 }
22
```

```
SpringRest/pom.xml | SpringRestApplication.java X
SpringRest > src/main/java > com.cts.SpringRest > SpringRestApplication > log: Logger
1 package com.cts.SpringRest;
2
Console X
<terminated> SpringRestApplication [Java Application] C:\Users\hemap.p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.25.0.1.v20251108-1451\jre\bin\javaw.exe (1 Jul 2026, 9:53:01 am - 9:53:12 am elapsed: 0:00:11.409) [pi
09:53:01.655 [main] INFO com.cts.SpringRest.SpringRestApplication -- Application is starting...
09:53:01.906 [restartedMain] INFO com.cts.SpringRest.SpringRestApplication -- Application is starting...

  ____ _
 / ___ \| | | |
/ /___ \| |_| |
 \___ \| __|_| |
  ___ \| | | |
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/ /___ \| |_| |
 \___ \| __|_| |

:: Spring Boot ::                (v3.5.3)

2026-07-01T09:53:02.380+05:30 INFO 18144 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Starting SpringRestApplication using Java 25.0.1 with PID :
2026-07-01T09:53:02.382+05:30 INFO 18144 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : No active profile set, falling back to 1 default profile: '
2026-07-01T09:53:02.457+05:30 INFO 18144 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.ad
2026-07-01T09:53:02.457+05:30 INFO 18144 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'l
2026-07-01T09:53:03.673+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-01T09:53:03.697+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T09:53:03.697+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2026-07-01T09:53:03.756+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-01T09:53:03.756+05:30 INFO 18144 --- [SpringRest] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 12
2026-07-01T09:53:04.190+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-01T09:53:04.229+05:30 INFO 18144 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-01T09:53:04.243+05:30 INFO 18144 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Started SpringRestApplication in 2.332 seconds (process run
2026-07-01T09:53:04.247+05:30 INFO 18144 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Application started successfully.
```

Exercise – 2 - Spring Core – Load Country from Spring Configuration XML

```
http://www.springframework.org/schema/beans/spring-beans.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2 <beans xmlns="http://www.springframework.org/schema/beans"
3     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4     xsi:schemaLocation="http://www.springframework.org/schema/beans
5         http://www.springframework.org/schema/beans/spring-beans.xsd">
6
7     <bean id="country" class="com.cts.SpringRest.country.Country">
8         <property name="code" value="IN" />
9         <property name="name" value="India" />
10    </bean>
11
12 </beans>
13
14
15
```

SpringRest/p... SpringRestAp... application... X cou

```
1 spring.application.name=SpringRest
2 logging.level.com.cts.SpringRest=DEBUG
```

```
SpringRest/p... SpringRestAp... application... country.xml Country.java X LibraryManag...
SpringRest > src/main/java > com.cts.SpringRest.country > Country >
1 package com.cts.SpringRest.country;
2
3
4 import lombok.extern.slf4j.Slf4j;
5
6 @Slf4j
7 public final class Country {
8     private String code;
9     private String name;
10
11     public Country() {
12         log.debug("Inside Country Constructor");
13     }
14
15     @Override
16     public String toString() {
17         return "Country [code=" + code + ", name=" + name + "]";
18     }
19
20     public String getCode() {
21         return code;
22     }
23     public void setCode(String code) {
24         this.code = code;
25     }
26     public String getName() {
27         return name;
28     }
29     public void setName(String name) {
30         this.name = name;
31     }
32
33 }
34
```


SpringRest/p...
SpringRestAp...
application...
country.xml
Country.java
LibraryManag...

SpringRest
src/main/java
com.cts.SpringRest
SpringRestApplication
displayCountry(): void

```

1 package com.cts.SpringRest;
2
3
4 import org.springframework.boot.SpringApplication;
5 import org.springframework.boot.autoconfigure.SpringBootApplication;
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7
8 import com.cts.SpringRest.country.Country;
9
10 import org.springframework.context.*;
11
12 //import com.cts.SpringRest.country.Country;
13
14 import lombok.extern.slf4j.Slf4j;
15
16 @Slf4j
17 @SpringBootApplication
18 public class SpringRestApplication {
19
20     public static void displayCountry() {
21         System.out.println("Inside displayCountry()");
22         ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
23         Country country = (Country) context.getBean("country", Country.class);
24         Log.debug("Country : {}", country.toString());
25     }
26
27     public static void main(String[] args) {
28
29         SpringApplication.run(SpringRestApplication.class);
30         System.out.println("Main method started");
31         displayCountry();
32     }
33
34
35

```

Console X
<terminated> SpringRestApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.just\openjdk.hotspot.jre.full.win32.x86_64_25.0.1.x20251108-1451\jre\bin\javaw.exe (1 Jul 2026, 11:07:03 am - 11:08:15 am elapsed: 0:01:11.986) [pid: 21756]

```

:: Spring Boot ::
(v3.5.3)

2026-07-01T11:07:05.465+05:30 INFO 21756 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Starting SpringRestApplication using Java 25.0.1 with PID 21756 (C:\Users\hemap\Down...
2026-07-01T11:07:05.468+05:30 DEBUG 21756 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Running with Spring Boot v3.5.3, Spring v6.2.8
2026-07-01T11:07:05.469+05:30 INFO 21756 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-01T11:07:05.551+05:30 INFO 21756 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
2026-07-01T11:07:05.552+05:30 INFO 21756 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-01T11:07:06.857+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-01T11:07:06.874+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T11:07:06.874+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2026-07-01T11:07:06.940+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-01T11:07:06.941+05:30 INFO 21756 --- [SpringRest] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1388 ms
2026-07-01T11:07:07.423+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-01T11:07:07.454+05:30 INFO 21756 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-01T11:07:07.467+05:30 INFO 21756 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Started SpringRestApplication in 2.507 seconds (process running for 2.954)

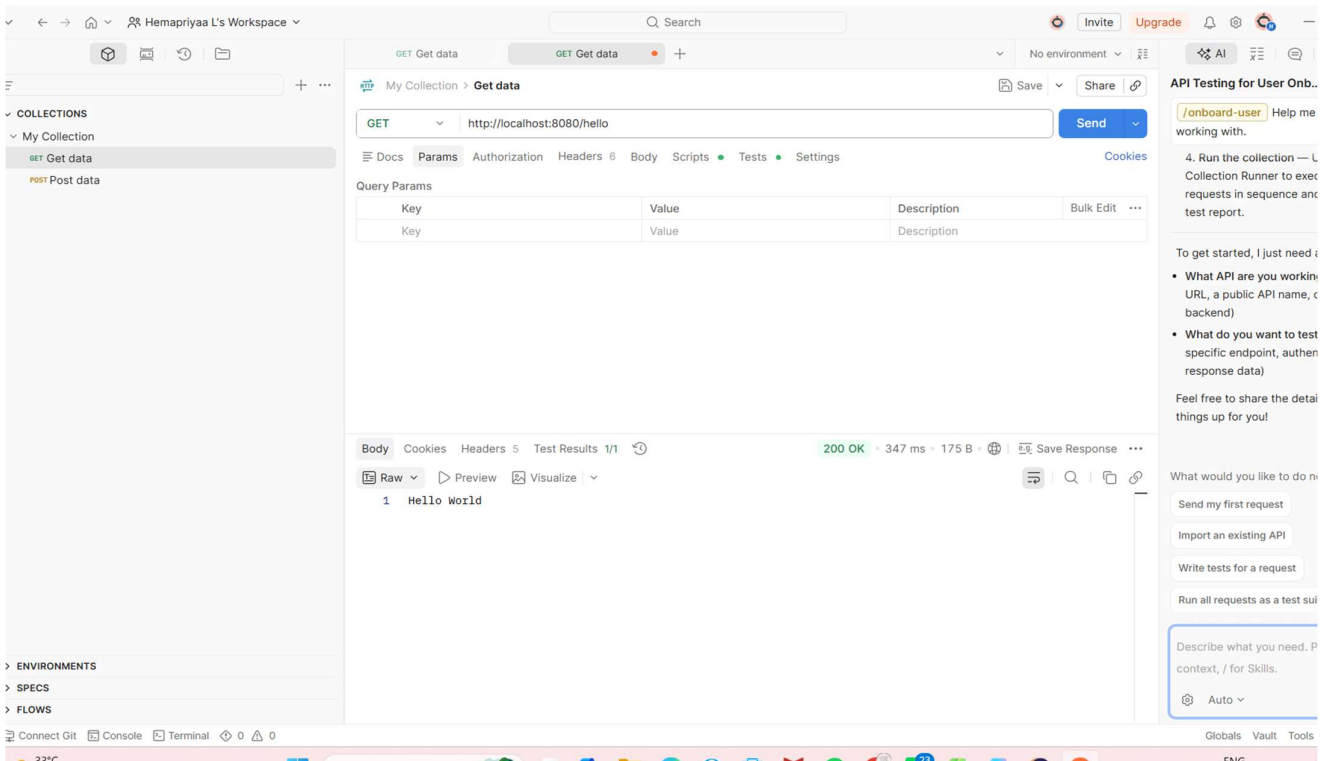
Main method started
Inside displayCountry()
2026-07-01T11:07:07.591+05:30 DEBUG 21756 --- [SpringRest] [ restartedMain] com.cts.SpringRest.country.Country : Inside Country Constructor
2026-07-01T11:07:07.595+05:30 DEBUG 21756 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Country : Country [code=IN, name=India]
2026-07-01T11:08:13.522+05:30 INFO 21756 --- [SpringRest] [ File Watcher] rtingClassPathChangeChangedEventListener : Restarting due to 1 class path change (0 additions, 0 deletions, 1 modification)
2026-07-01T11:08:13.537+05:30 INFO 21756 --- [SpringRest] [ Thread-5] o.s.b.w.e.tomcat.GracefulShutdown : Commencing graceful shutdown. Waiting for active requests to complete
2026-07-01T11:08:13.547+05:30 INFO 21756 --- [SpringRest] [ tomcat-shutdown] o.s.b.w.e.tomcat.GracefulShutdown : Graceful shutdown complete

```

Ex – 3 -Hello World RESTful Web Service

```
Ques73.java  HelloController.java  *SpringRestApplication.java
SpringRest ▶ src/main/java ▶ com.cts.SpringRest.controller ▶ HelloController ▶ sayHello() : String
1  package com.cts.SpringRest.controller;
2
3  import org.springframework.web.bind.annotation.GetMapping;
4  import org.springframework.web.bind.annotation.RestController;
5
6  import lombok.extern.slf4j.Slf4j;
7
8  @Slf4j
9  @RestController
10 public class HelloController {
11
12     @GetMapping("/hello")
13     public String sayHello() {
14         log.info("START - in sayHelloMethod");
15         String msg = "Hello World";
16         log.info("END - in sayHelloMethod");
17         return msg;
18     }
19 }
20
```

```
SpringRest ▶ src/main/java ▶ com.cts.SpringRest ▶ SpringRestApplication ▶
1  package com.cts.SpringRest;
2
3
4  import org.springframework.boot.SpringApplication;
5
6
7  @Slf4j
8  @SpringBootApplication
9  public class SpringRestApplication {
10
11     public static void main(String[] args) {
12         SpringApplication.run(SpringRestApplication.class);
13     }
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
```



```
Console X
SpringRestApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_25.0.1.v20251108-1451\jre\bin\javaw.exe (1 Jul 2026, 6:42:01 pm elapsed: 0:01:47) [pid: 26932]

:: Spring Boot :: (v3.5.3)

2026-07-01T18:42:04.116+05:30 INFO 26932 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Starting SpringRestApplication using Java 25.0.1 with PID 269
2026-07-01T18:42:04.124+05:30 DEBUG 26932 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Running with Spring Boot v3.5.3, Spring v6.2.8
2026-07-01T18:42:04.126+05:30 INFO 26932 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : No active profile set, falling back to 1 default profile: "d
2026-07-01T18:42:04.270+05:30 INFO 26932 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-i
2026-07-01T18:42:04.271+05:30 INFO 26932 --- [SpringRest] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'log
2026-07-01T18:42:06.607+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-01T18:42:06.644+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-01T18:42:06.645+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2026-07-01T18:42:06.743+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-01T18:42:06.743+05:30 INFO 26932 --- [SpringRest] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2469
2026-07-01T18:42:07.657+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-01T18:42:07.717+05:30 INFO 26932 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-01T18:42:07.736+05:30 INFO 26932 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Started SpringRestApplication in 4.417 seconds (process runn
2026-07-01T18:42:52.513+05:30 INFO 26932 --- [SpringRest] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-01T18:42:52.513+05:30 INFO 26932 --- [SpringRest] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-01T18:42:52.518+05:30 INFO 26932 --- [SpringRest] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms
2026-07-01T18:42:52.622+05:30 INFO 26932 --- [SpringRest] [nio-8080-exec-1] c.c.s.controller.HelloController : START - in sayHelloMethod
2026-07-01T18:42:52.622+05:30 INFO 26932 --- [SpringRest] [nio-8080-exec-1] c.c.s.controller.HelloController : END - in sayHelloMethod
```

Ex – 4 - REST - Country Web Service

```
plorer X
Ques73.java HelloController.java SpringRestApplication.java CountryController.java Spring

st/java
4
t
md
/
i.cmd
xml
ava
est
ain/java
m.cts.SpringRest
SpringRestApplication.java
m.cts.SpringRest.controller

http://www.springframework.org/schema/beans/spring-beans.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2 <beans xmlns="http://www.springframework.org/schema/beans"
3   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4   xsi:schemaLocation="http://www.springframework.org/schema/beans
5     http://www.springframework.org/schema/beans/spring-beans.xsd">
6
7   <bean id="country" class="com.cts.SpringRest.country.Country">
8     <property name="code" value="IN" />
9     <property name="name" value="India" />
10  </bean>
11
12
13 </beans>
14
15
16
```



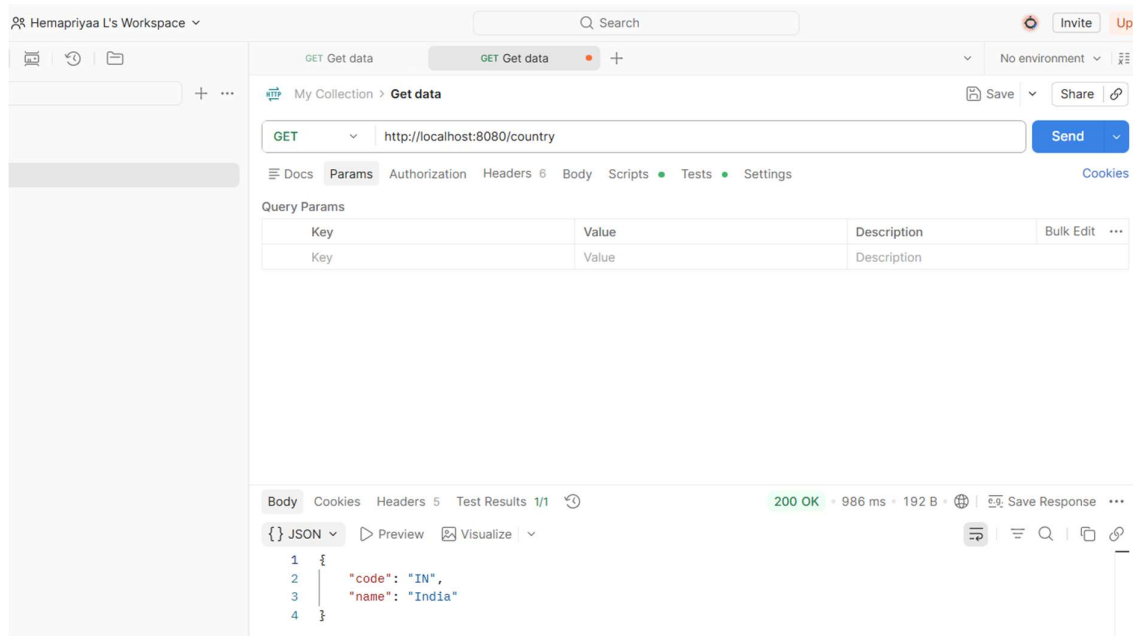
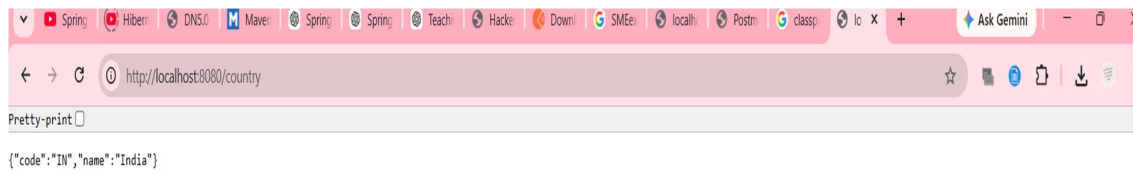
```
Ques73.java HelloController.java SpringRestApplication.java CountryController.java
SpringRest > src/main/java > com.cts.SpringRest.controller > CountryController
1 package com.cts.SpringRest.controller;
2
3 import java.util.List;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6 import org.springframework.web.bind.annotation.GetMapping;
7 import org.springframework.web.bind.annotation.RequestMapping;
8 import org.springframework.web.bind.annotation.RestController;
9
10 import com.cts.SpringRest.country.Country;
11
12 @RequestMapping
13 @RestController
14 public class CountryController {
15
16     @GetMapping("/country")
17     public Country getCountryIndia()
18     {
19         ApplicationContext context =
20             new ClassPathXmlApplicationContext("country.xml");
21
22         Country country = context.getBean("country", Country.class);
23         return country;
24     }
25
26 }
```

```
SpringRest > src/main/java > com.cts.SpringRest > SpringRestApplication
1 package com.cts.SpringRest;
2
3
4 import org.springframework.boot.SpringApplication;
5
15
16 @Slf4j
17 @SpringBootApplication
18 public class SpringRestApplication {
19
20     public static void main(String[] args) {
21         SpringApplication.run(SpringRestApplication.class);
22     }
23
24
25
26
27
28
```

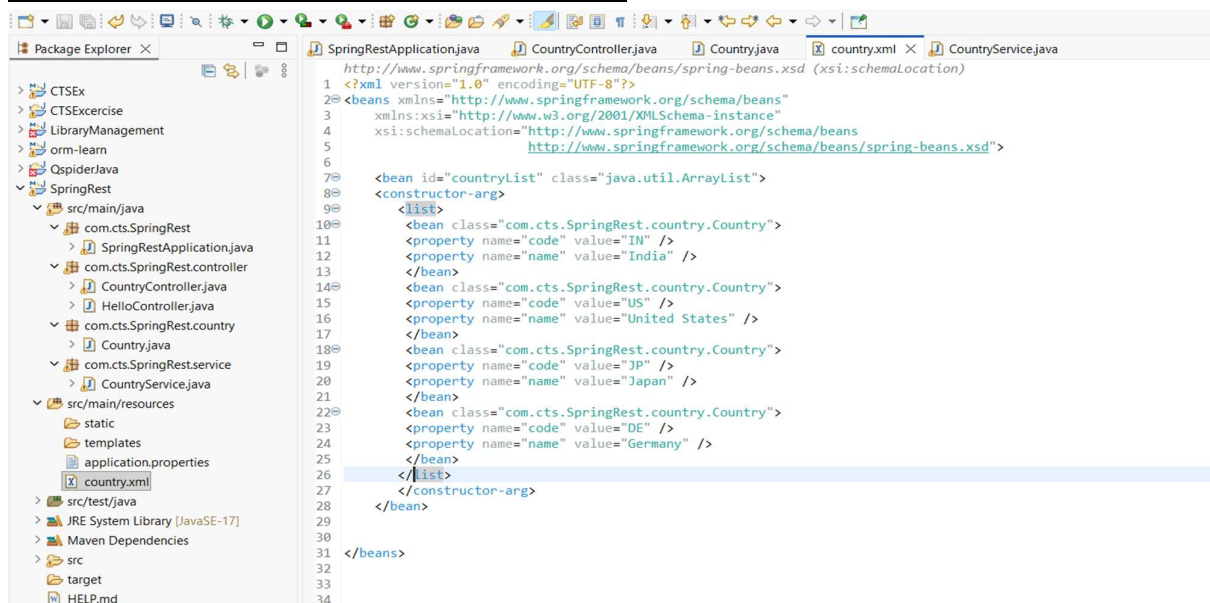
```
Console
SpringRestApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_25.0.1.v20251108-1451\jre\bin\javaw.exe (1 Jul 2026, 7:16:40 pm elapsed: 0:00:26) [pid: 25864]

:: Spring Boot :: (v3.5.3)

2026-07-01T19:16:41.503+05:30 INFO 25864 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Starting SpringRestApplication using Java 25.0.1 with PID 25864 (C:\Users\hemap\Down
2026-07-01T19:16:41.511+05:30 DEBUG 25864 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Running with Spring Boot v3.5.3, Spring v6.2.8
2026-07-01T19:16:41.514+05:30 INFO 25864 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-01T19:16:41.612+05:30 INFO 25864 --- [SpringRest] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
2026-07-01T19:16:41.612+05:30 INFO 25864 --- [SpringRest] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-01T19:16:44.130+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-01T19:16:44.166+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting service [Tomcat]
2026-07-01T19:16:44.168+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2026-07-01T19:16:44.223+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-01T19:16:44.225+05:30 INFO 25864 --- [SpringRest] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2610 ms
2026-07-01T19:16:45.256+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-07-01T19:16:45.356+05:30 INFO 25864 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-01T19:16:45.395+05:30 INFO 25864 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Started SpringRestApplication in 4.484 seconds (process running for 4.996)
2026-07-01T19:16:54.444+05:30 INFO 25864 --- [SpringRest] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-01T19:16:54.445+05:30 INFO 25864 --- [SpringRest] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-01T19:16:54.448+05:30 INFO 25864 --- [SpringRest] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 3 ms
2026-07-01T19:16:54.890+05:30 DEBUG 25864 --- [SpringRest] [nio-8080-exec-1] com.cts.SpringRest.country.Country : Inside Country Constructor
```



EX – 5 - REST - Get country based on country code



Package Explorer

- CTSEx
- CTSExercise
- LibraryManagement
- orm-learn
- QspiderJava
- SpringRest
 - src/main/java
 - com.cts.SpringRest
 - SpringRestApplication.java
 - com.cts.SpringRest.controller
 - CountryController.java
 - HelloController.java
 - com.cts.SpringRest.country
 - Country.java
 - com.cts.SpringRest.service
 - CountryService.java
 - src/main/resources
 - static
 - templates
 - application.properties
 - country.xml
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw

SpringRestApplication.java CountryController.java Country.java CountryService.java

```
1 package com.cts.SpringRest.country;
2
3
4 import lombok.extern.slf4j.Slf4j;
5
6 @Slf4j
7 public final class Country {
8     private String code;
9     private String name;
10
11     public Country() {
12         log.debug("Inside Country Constructor");
13     }
14
15     @Override
16     public String toString() {
17         return "Country [code=" + code + ", name=" + name + "]";
18     }
19
20     public String getCode() {
21         return code;
22     }
23     public void setCode(String code) {
24         this.code = code;
25     }
26     public String getName() {
27         return name;
28     }
29     public void setName(String name) {
30         this.name = name;
31     }
32 }
33
34
```

Package Explorer

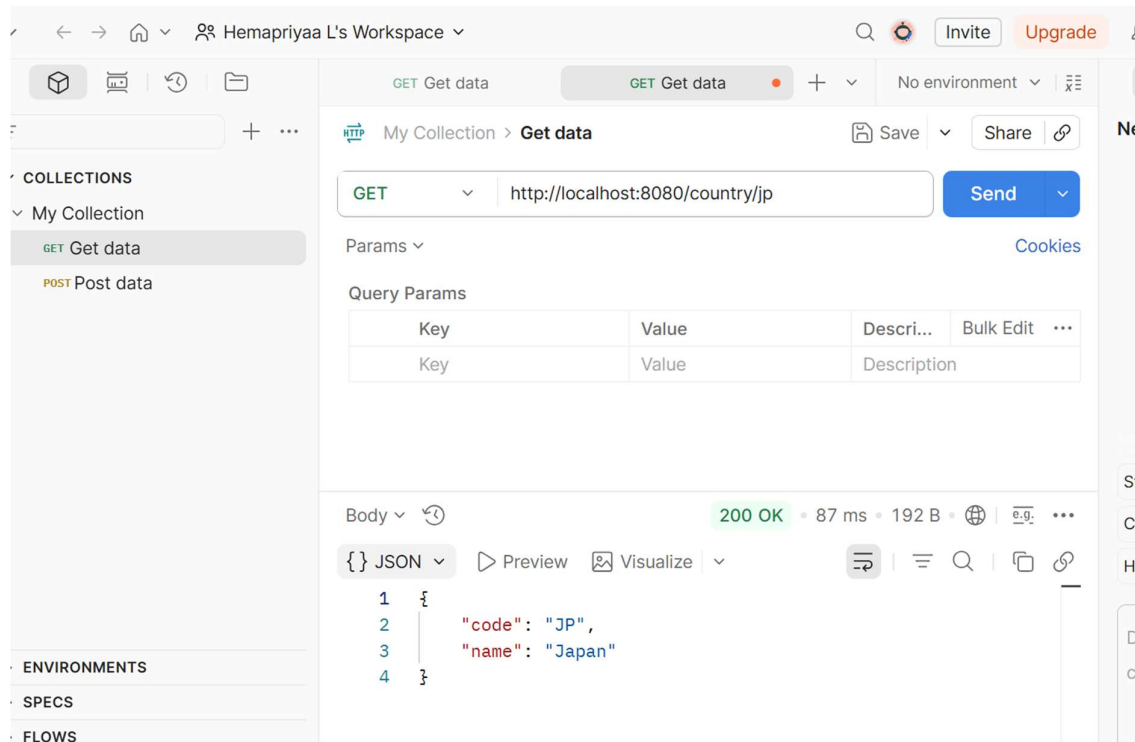
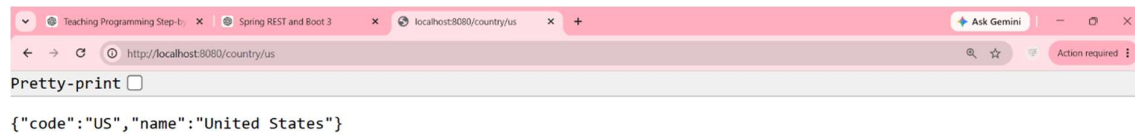
- CTSEx
- CTSExercise
- LibraryManagement
- orm-learn
- QspiderJava
- SpringRest
 - src/main/java
 - com.cts.SpringRest
 - SpringRestApplication.java
 - com.cts.SpringRest.controller
 - CountryController.java
 - HelloController.java
 - com.cts.SpringRest.country
 - Country.java
 - com.cts.SpringRest.service
 - CountryService.java
 - src/main/resources
 - static
 - templates
 - application.properties
 - country.xml
 - src/test/java
 - JRE System Library [JavaSE-17]
 - Maven Dependencies
 - src

SpringRestApplication.java CountryController.java CountryService.java

```
1 package com.cts.SpringRest.controller;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8 import org.springframework.web.bind.annotation.GetMapping;
9 import org.springframework.web.bind.annotation.*;
10
11 import com.cts.SpringRest.country.Country;
12 import com.cts.SpringRest.service.CountryService;
13
14 @RestController
15 @RequestMapping("/country")
16
17 public class CountryController {
18     @Autowired
19     private CountryService countryService;
20
21     @GetMapping("/{code}")
22     public Country getCountry(@PathVariable String code)
23     {
24
25         return countryService.getCountry(code);
26     }
27
28 }
29
```

```
1 package com.cts.SpringRest.service;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 import org.springframework.context.ApplicationContext;
7 import org.springframework.context.support.ClassPathXmlApplicationContext;
8 import org.springframework.stereotype.Service;
9
10 import com.cts.SpringRest.country.Country;
11
12 @Service
13 public class CountryService {
14
15     public Country getCountry(String code) {
16         ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
17         List<Country> countries = (List<Country>) context.getBean("countryList", ArrayList.class);
18
19         for( Country x : countries )
20         {
21             if(x.getCode().equalsIgnoreCase(code))
22                 return x;
23         }
24
25         return null;
26     }
27 }
28
29
30
```

```
1 package com.cts.SpringRest;
2
3
4 import org.springframework.boot.SpringApplication;
5
6 @Slf4j
7 @SpringBootApplication
8 public class SpringRestApplication {
9
10     public static void main(String[] args) {
11         SpringApplication.run(SpringRestApplication.class);
12     }
13 }
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
```

Ex – 6- Create authentication service that returns JWT




```
1 package com.cts.SpringRest.util;
2
3 import java.security.Key;
4 import java.util.Date;
5
6 import io.jsonwebtoken.Jwts;
7 import io.jsonwebtoken.SignatureAlgorithm;
8 import io.jsonwebtoken.security.Keys;
9
10 import org.springframework.stereotype.Component;
11
12 @Component
13 public class JwtUtil {
14     private static final String SECRET =
15         "mysecretkeymysecretkeymysecretkey12";
16
17     private Key getKey() {
18         return Keys.hmacShaKeyFor(SECRET.getBytes());
19     }
20
21     public String generateToken(String username) {
22
23         return Jwts.builder()
24             .setSubject(username)
25             .setIssuedAt(new Date())
26             .setExpiration(new Date(System.currentTimeMillis() + 1200000))
27             .signWith(getKey(), SignatureAlgorithm.HS256)
28             .compact();
29     }
30 }
31 }
```

```
1 package com.cts.SpringRest.model;
2
3 public class AuthenticationResponse {
4     private String token;
5
6     public AuthenticationResponse() {
7     }
8
9     public AuthenticationResponse(String token) {
10         this.token = token;
11     }
12
13     public String getToken() {
14         return token;
15     }
16
17     public void setToken(String token) {
18         this.token = token;
19     }
20 }
21 }
```

Package Explorer

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- SpringRest
 - src/main/java
 - com.cts.SpringRest
 - SpringRestApplication.java
 - com.cts.SpringRest.config
 - SecurityConfig.java
 - com.cts.SpringRest.controller
 - AuthenticateController.java
 - CountryController.java
 - HelloController.java
 - com.cts.SpringRest.countr
 - com.cts.SpringRest.model
 - AuthenticationResponse
 - com.cts.SpringRest.service
 - com.cts.SpringRest.util
 - JwtUtil.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-1
 - Maven Dependencies
 - src
 - target
 - HELP.md
 - mvnw
 - mvnw.cmd

AuthenticateController.java

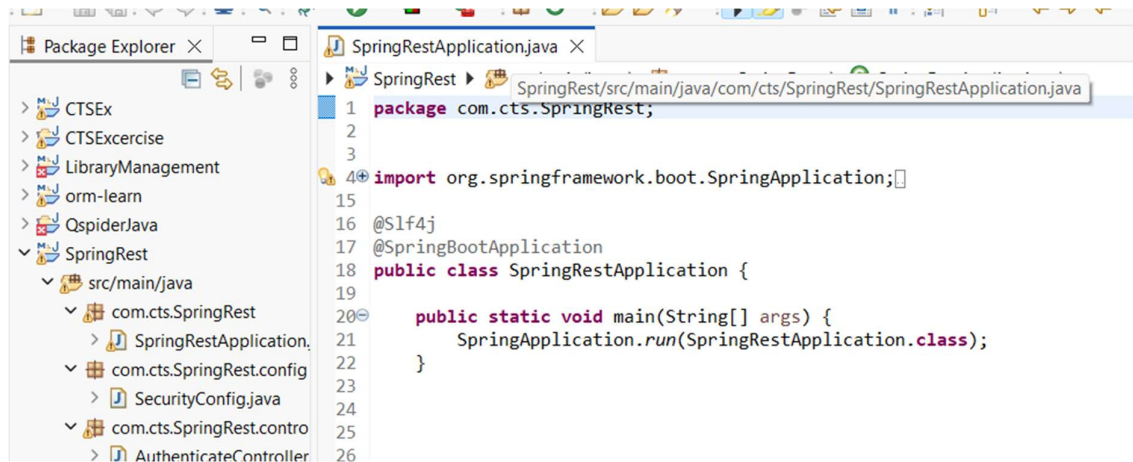
```
1 package com.cts.SpringRest.controller;
2
3 import java.nio.charset.StandardCharsets;
4 import java.util.Base64;
5
6 import org.springframework.beans.factory.annotation.Autowired;
7 import org.springframework.http.HttpHeaders;
8 import org.springframework.web.bind.annotation.*;
9
10 import com.cts.SpringRest.model.AuthenticationResponse;
11 import com.cts.SpringRest.util.JwtUtil;
12
13 @RestController
14 public class AuthenticateController {
15     @Autowired
16     private JwtUtil jwtUtil;
17
18     @GetMapping("/authenticate")
19     public AuthenticationResponse authenticate(
20         @RequestHeader(HttpHeaders.AUTHORIZATION) String authorizationHeader) {
21
22         System.out.println("Inside authenticate()");
23
24         String encoded = authorizationHeader.substring(6);
25
26         byte[] decodedBytes = Base64.getDecoder().decode(encoded);
27
28         String credentials = new String(decodedBytes, StandardCharsets.UTF_8);
29
30         String username = credentials.split(":")[0];
31
32         String token = jwtUtil.generateToken(username);
33
34         return new AuthenticationResponse(token);
35     }
36 }
37
```

Package Explorer

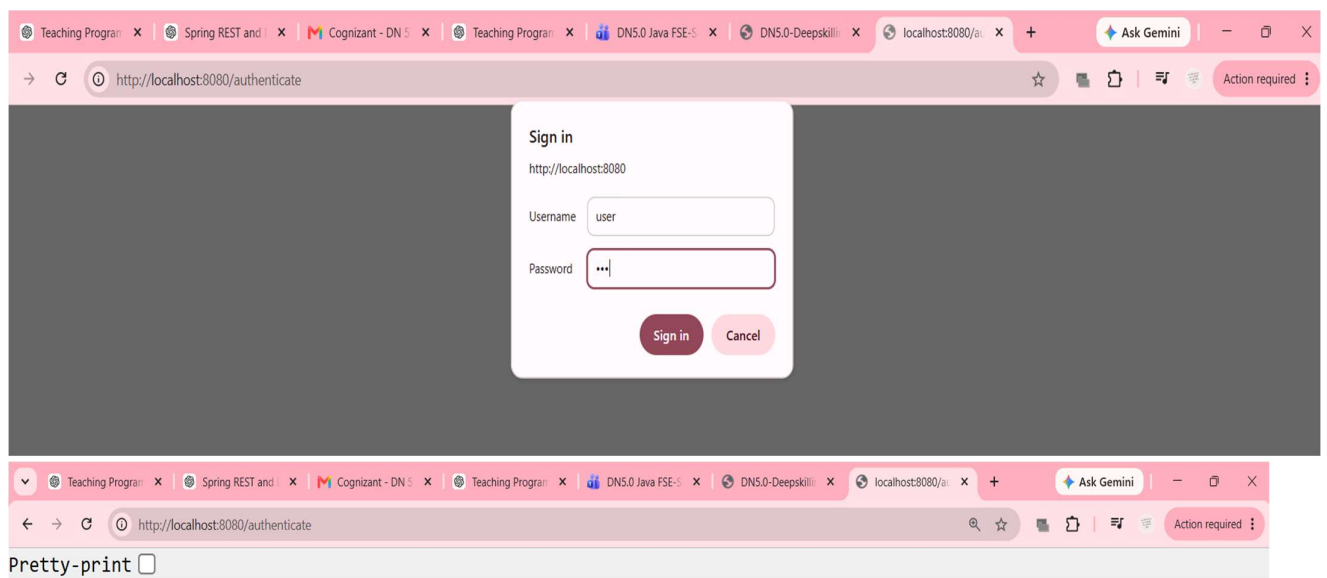
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 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml

*SecurityConfig.java

```
1 package com.cts.SpringRest.config;
2
3 import org.springframework.context.annotation.Bean;
4 import org.springframework.context.annotation.Configuration;
5 import org.springframework.security.config.Customizer;
6 import org.springframework.security.config.annotation.web.builders.HttpSecurity;
7 import org.springframework.security.core.userdetails.User;
8 import org.springframework.security.core.userdetails.UserDetails;
9 import org.springframework.security.provisioning.InMemoryUserDetailsManager;
10 import org.springframework.security.web.SecurityFilterChain;
11
12 @Configuration
13 public class SecurityConfig {
14
15     @Bean
16     public InMemoryUserDetailsManager userDetailsService() {
17
18         UserDetails user = User
19             .withUsername("user")
20             .password("{noop}pwd")
21             .roles("USER")
22             .build();
23
24         return new InMemoryUserDetailsManager(user);
25     }
26
27     @Bean
28     public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
29
30         http
31             .csrf(csrf -> csrf.disable())
32             .authorizeHttpRequests(auth -> auth
33                 .requestMatchers("/authenticate").authenticated()
34                 .anyRequest().permitAll())
35             .httpBasic(Customizer.withDefaults());
36
37         return http.build();
38     }
39 }
40
```



```
1 package com.cts.SpringRest;
2
3
4 import org.springframework.boot.SpringApplication;
5
6
7 @Slf4j
8 @SpringBootApplication
9 public class SpringRestApplication {
10
11     public static void main(String[] args) {
12         SpringApplication.run(SpringRestApplication.class);
13     }
14 }
```



```
{"token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ1c2VyIiwiaWF0IjoxNzgyOTkxNzI2LCJleHAiOiJlZ3ODI5OTI5MjZ9.KZ1b2F4Hm5WqmR01UVr-LujRj0nQ23Jm_c_X3Yt5-aM"}
```

Hemapriyaa L's Workspace

Search

Invite Upgrade

GET Get data

My Collection > Get data

Save Share

GET http://localhost:8080/authenticate Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

Auth Type

Basic Auth

Username user

Password pwd

The authorization header will be automatically generated when you send the request. Learn more about [Basic Auth](#) authorization.

Body

200 OK • 310 ms • 475 B • Save Response

JSON Preview Visualize

```
{
  "token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ1c2VyIiwiaWF0IjoxNzgyOTkyMzExLCJleHAiOjE3ODI5OTM1MTF9.9CmcqwcK2dMRzVmQnp7zIxUVdU9I2lhuUFimCcarc9Y"
}
```

```
SpringRestApplication [Java Application] C:\Users\hemapriyaa\p2\pool\plugins\org.eclipse.jdt.launcher\org.eclipse.jdt.launcher.exe (2 Jul 2026, 5:04:36 pm elapsed: 0:04:29) [pid: 26500]

:: Spring Boot :: (v3.5.3)

2026-07-02T17:04:37.519+05:30 INFO 26500 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Starting SpringRestApplication using Java 25.0.1 with PID 26500 (C:\Users\hemapriyaa\p2\pool\plugins\org.eclipse.jdt.launcher\org.eclipse.jdt.launcher.exe)
2026-07-02T17:04:37.522+05:30 DEBUG 26500 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Running with Spring Boot v3.5.3, Spring v6.2.8
2026-07-02T17:04:37.523+05:30 INFO 26500 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-02T17:04:37.588+05:30 INFO 26500 --- [SpringRest] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-02T17:04:37.588+05:30 INFO 26500 --- [SpringRest] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-02T17:04:38.865+05:30 INFO 26500 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-02T17:04:38.880+05:30 INFO 26500 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-02T17:04:38.880+05:30 INFO 26500 --- [SpringRest] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2026-07-02T17:04:38.919+05:30 INFO 26500 --- [SpringRest] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-07-02T17:04:39.056+05:30 INFO 26500 --- [SpringRest] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1331 ms
2026-07-02T17:04:39.456+05:30 INFO 26500 --- [SpringRest] [ restartedMain] r$InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsServiceImpl bean with name userDetailsService
2026-07-02T17:04:39.490+05:30 INFO 26500 --- [SpringRest] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : LiveReload server is running on port 35729
2026-07-02T17:04:39.501+05:30 INFO 26500 --- [SpringRest] [ restartedMain] c.cts.SpringRest.SpringRestApplication : Tomcat started on port 8080 (http) with context path '/'
2026-07-02T17:04:45.752+05:30 INFO 26500 --- [SpringRest] [nio-8080-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/] : Started SpringRestApplication in 2.348 seconds (process running for 2.663)
2026-07-02T17:04:45.752+05:30 INFO 26500 --- [SpringRest] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-02T17:04:45.753+05:30 INFO 26500 --- [SpringRest] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-02T17:04:45.753+05:30 INFO 26500 --- [SpringRest] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
Inside authenticate()
```